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STATISTICS HSSC–I
SECTION – A (Marks 17)
Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.
Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Version No.				
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ROLL NUMBER

Answer Sheet No. ____________

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔ **Invigilator Sign.** ____________

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** ____________

	Question سوال	A	B	C	D	A	B	C	D
1.	Numerical characteristic/value calculated from population is called:	Constant	Samples	Parameter	Statistic	○	○	○	○
2.	The difference between upper class boundary and lower class boundary is called:	Class frequency	Class limit	Class mark	Class size	○	○	○	○
3.	The process of dividing the data into different groups according to some common characteristics is called:	Classification	Tabulation	Frequency distribution	Array	○	○	○	○
4.	Height measurements of students is a type of:	Discrete variable	Continuous variable	Factious variable	Qualitative variable	○	○	○	○
5.	In a symmetrical frequency distribution, if $Q_1 = 7$ and $Q_3 = 19$ then median will be:	12	13	15	18	○	○	○	○
6.	If minimum value is 10 and maximum value is 30 in a series of data then the value of coefficient of range is:	20	40	0.50	0.60	○	○	○	○
7.	For a series of data of variable X, if $\overline{X} = 5$ and $Y = 3X - 6$, then $\overline{Y} =$	10	9	7	5	○	○	○	○
8.	The lack of symmetry in frequency distribution is called:	Dispersion	Skewness	Kurtosis	Variation	○	○	○	○
9.	Percentage of values lies between interval $\overline{X} \pm 3S$ in a normal distribution is:	50%	68.27%	95.45%	99.73%	○	○	○	○
10.	If Var(x) = 2 and Var (y) = 3 then Var (X-Y)=	-1	Zero	5	13	○	○	○	○
11.	If all values considered in calculating an index number are of equal importance, then index number is called:	Weighted index number	Un-weighted index number	Simple index number	Composite index number	○	○	○	○

	Question سوال	A	B	C	D	A	B	C	D
12.	When index number is calculated for several variables, it is called:	Simple index number	Composite index number	Consumer price index number	Simple aggregative price index number	○	○	○	○
13.	In a method of least square the sum of squares of residuals is always:	Positive	Zero	Maximum	Minimum	○	○	○	○
14.	If $\hat{Y} = 4 + 0.95X$, then the value of regression coefficient Y on X is:	4	8	0.95	Zero	○	○	○	○
15.	The formula for correlation coefficient (r) is:	$\frac{S_{yx}}{S_x.S_y}$	$\frac{S_{yx}}{\sqrt{S_x.S_y}}$	$\frac{S_{yx}}{\sqrt{S_x^2}}$	$\frac{S_{yx}}{\sqrt{S_y^2}}$	○	○	○	○
16.	Suppose that $Y = 20 + 5X$ describes the an annual time series for year 1992-2000 with origin at 1996, then what will be the estimated trend value for 1996?	5	15	20	25	○	○	○	○
17.	Long term variation in the time series is regarded as:	Random variation	Secular variation	Cyclical variation	Seasonal variation	○	○	○	○

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STATISTICS HSSC–I

Time allowed: 2:35 Hours **Total Marks Sections B and C: 68**

SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. **(14 x 3 = 42)**

(i) Differentiate between discrete and continuous variable.

(ii) What methods are used for collection of primary data?

(iii) Read the following paragraph and make the frequency distribution of number of letters in each word with one as class interval.
"Studying books is the main source of knowledge. Books are indeed never failing friends of man. For a mature mind, reading is the greatest source of pleasure and solace to distressed minds".

(iv) A student obtained the following marks in four subjects. Compute weighted arithmetic mean.

Subject	English	Urdu	Physics	Chemistry
Marks (X)	50	60	80	90
Weight (W)	2	1	4	3

(v) Calculate arithmetic mean of a frequency distribution of X if $U = \frac{X-600}{10}, \sum fU = -20, \sum f = 5$

(vi) Compute Geometric and Harmonic mean of: 10,15,20,30

(vii) The following is a frequency distribution of the number of children born in 65 families. Find median and mode for this data.

No. of children (X)	0	1	2	3	4	5
No. of families (f)	4	7	14	22	12	6

(viii) If lower quartile is 20 and upper quartile is 40. Calculate semi-inter quartile range and coefficient of quartile deviation.

(ix) A student calculated mean and variance of 25 observations as 20 and 16 respectively. Compute coefficient of variation.

(x) If Median=20 and Arithmetic mean =22, compute the value of mode.

(xi) Calculate mean deviation from mean and mean coefficient of dispersion of 6, 8,10,12,14.

(xii) Compute price index numbers taking "average of first 03 years" as base:

Year	2000	2001	2002	2003	2004	2005	2006
Price	20	15	25	28	40	50	60

(xiii) Given that: $\Sigma p_1q_o = 9925,\ \Sigma p_oq_o = 9270,\ \Sigma p_1q_1 = 10710,\ \Sigma p_oq_1 = 10000$. Compute Laspyres, Paasche and Fisher Price index numbers.

(xiv) Differentiate between simple and composite index number.

(xv) If $r_{yx} = 0.97$ and $b_{xy} = 0.82$, then calculate b_{yx}

(xvi) If $S_x = 10$, $S_y = 8$ and $r_{yx} = 0.75$, then compute two regression coefficients Y on X and X on Y.

(xvii) Describes the properties of correlation coefficient (r).

(xviii) If the second degree parabola fitted to the data for the years 1980-87 (both inclusive) with the origin at the middle of 1983 and 1984 is $\hat{Y} = 131.81 + 16.89X + 1.64X^2$, the unit of X being half year, then find the trend values for 1980-87.

(xix) What is meant by irregular or random variation in time series?

SECTION – C (Marks 26)

Note: Attempt any TWO questions. **(2 x 13 = 26)**

Q. 3 The following table shows the marks obtained by the students:

Marks	10 -19	20 – 29	30 – 39	40 - 49	50 – 59	60 - 69
f	5	8	10	12	9	6

(i) Calculate the arithmetic mean, median and mode. **(06)**

(ii) Compute variance, standard deviation and coefficient of skewness. **(07)**

Q. 4 a. Compute index numbers of prices from the following data taking 1970 as base using Arithmetic Mean as an average. **(07)**

Year	Average price in Rupees		
	Firewood	Soft coke	Kerosene oil
1970	3.25	2.50	0.20
1971	3.44	2.80	0.22
1972	3.50	2.00	0.25
1973	3.75	2.50	0.30

b. Find regression line Y on X and correlation coefficient (r): **(06)**

$n = 8, \sum X = 36, \sum Y = 152, \sum XY = 768, \sum X^2 = 204, \sum Y^2 = 3056$

Q. 5 a. Compute (i) 5-years simple moving average (ii) 4 years centered moving average. **(06)**

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Values	20	23	26	29	23	29	31	27	38	34	33

b Fit a straight line to the following data and calculate trend values. **(07)**

Year	2001	2002	2003	2004	2005	2006	2007	2008	2009
Values	60	65	80	73	97	105	93	111	117